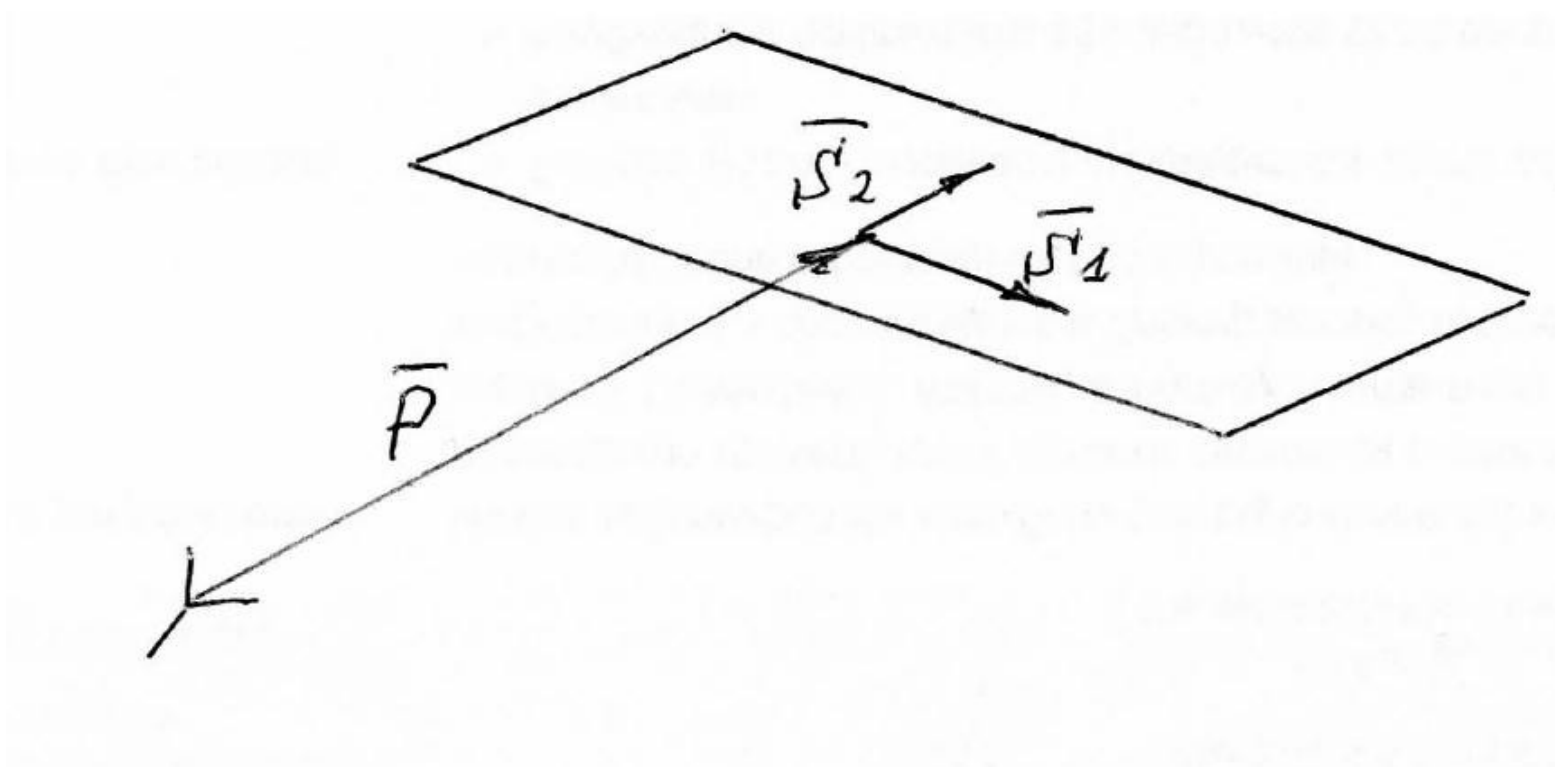


Типы поверхностей

Бесконечная плоскость

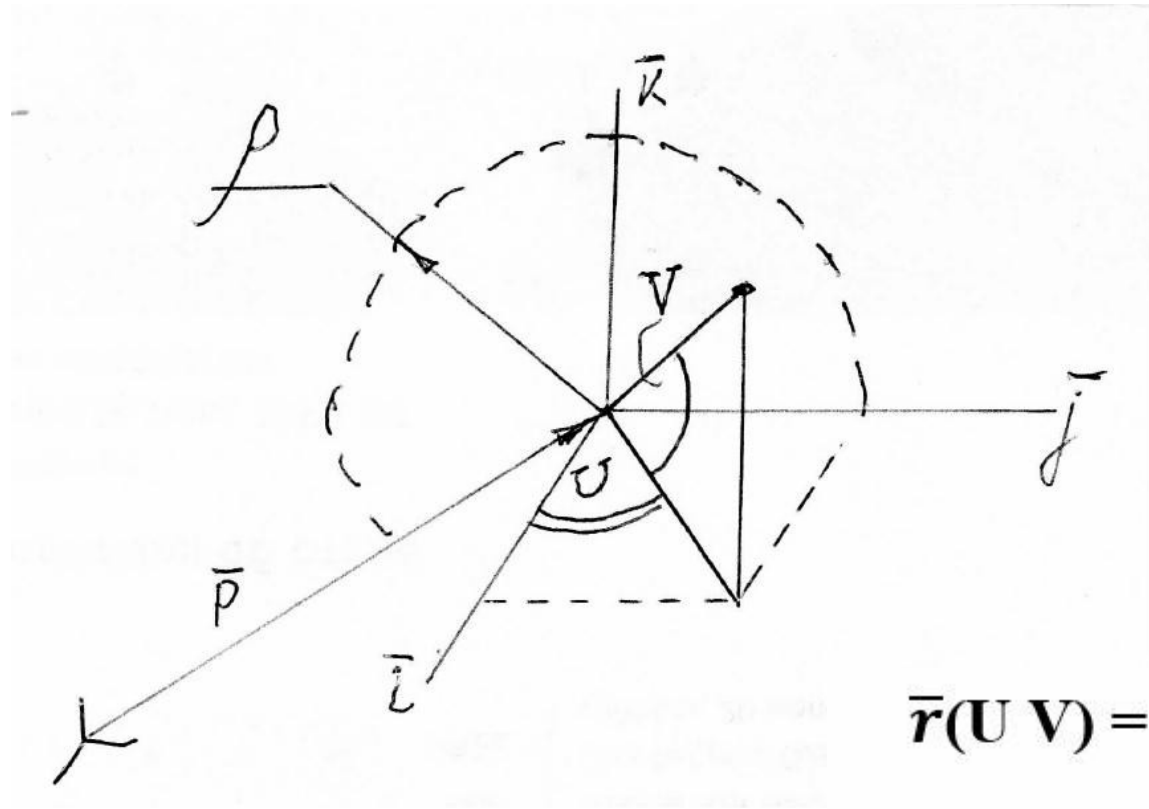


$$\overline{S(UV)} = \overline{P} + \overline{S_1} * U + \overline{S_2} * V$$

$$-\infty \leq U \leq \infty$$

$$-\infty \leq V \leq \infty$$

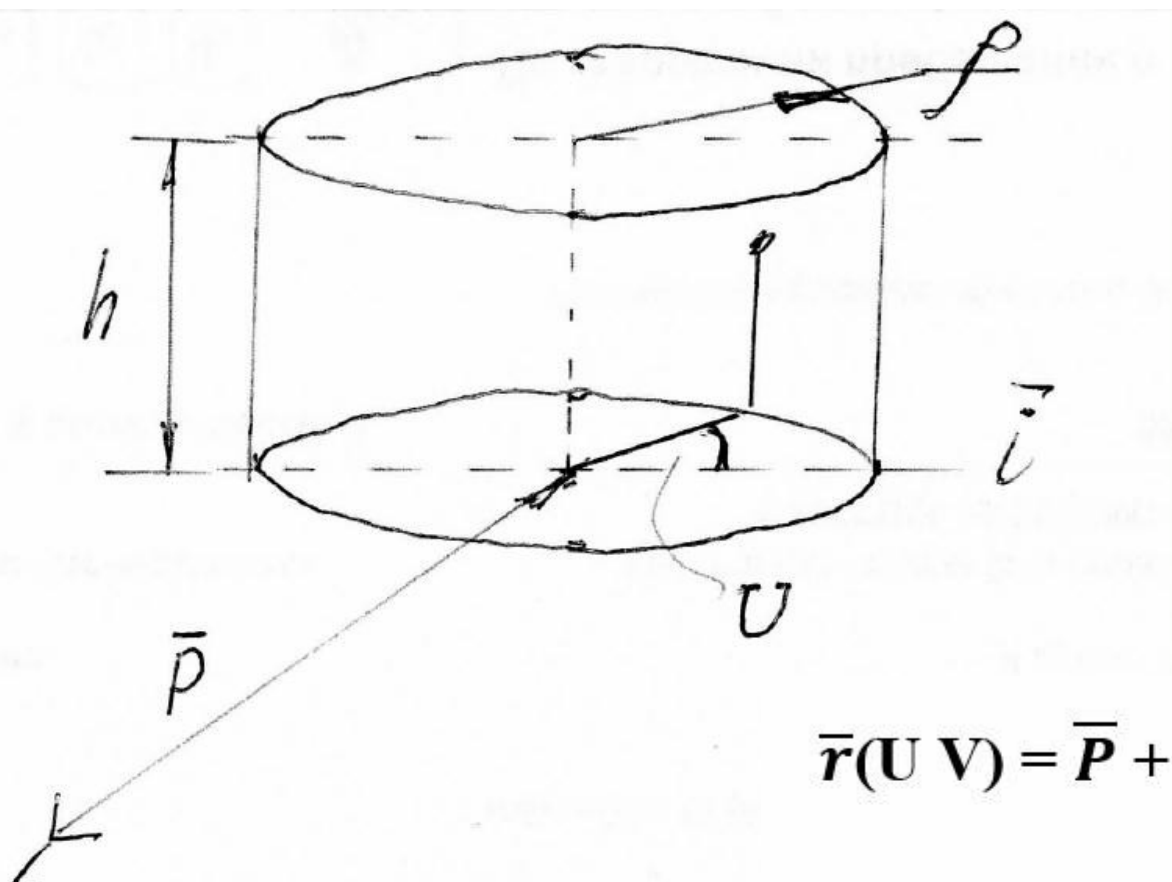
Сферическая поверхность



$$\vec{r}(U, V) = \vec{P} + \rho * \cos V * \cos U * \vec{i} + \rho * \cos V * \sin U * \vec{j} + \rho * \sin V * \vec{k} ;$$

$$0 \leq U \leq 2\pi \quad -\frac{\pi}{2} \leq V \leq \frac{\pi}{2}$$

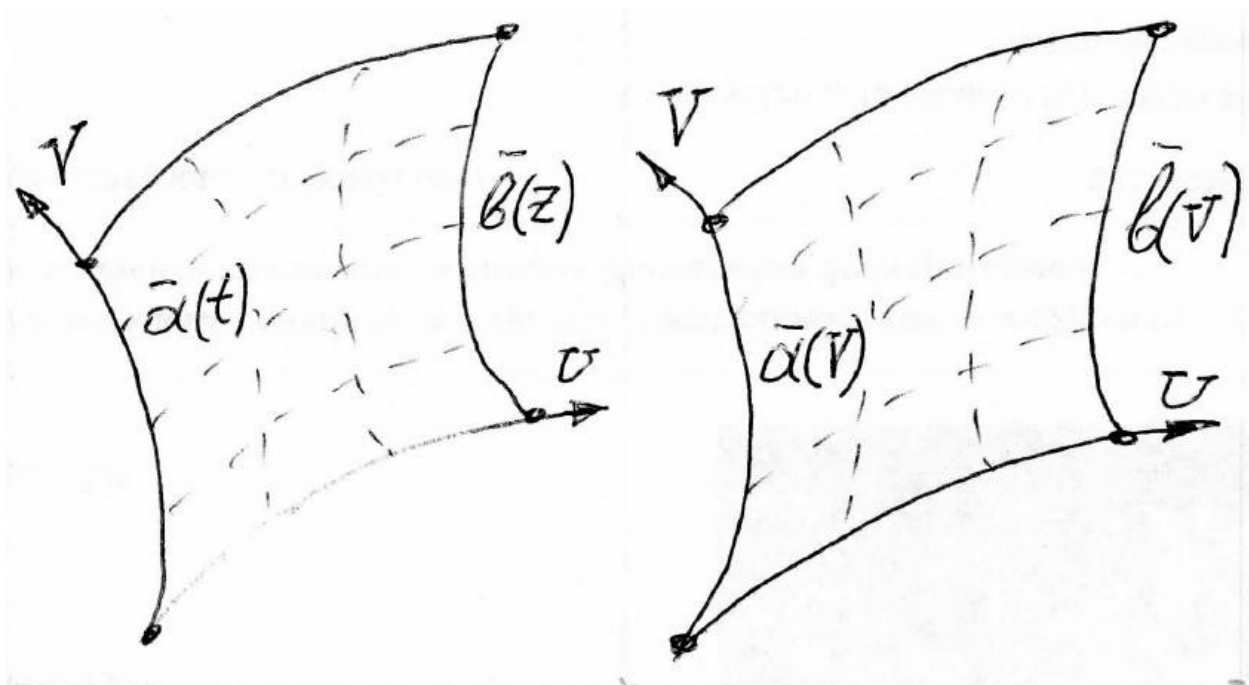
Цилиндрическая поверхность



$$\bar{r}(U, V) = \bar{P} + \rho * \cos U * \bar{i} + \rho * \sin U * \bar{j} + h * V * \bar{k} ;$$

$$0 \leq U \leq 2\pi \quad 0 \leq V \leq 1$$

Приведение уравнений противоположных ребер фрагмента поверхности к одному параметру



$$t_{\min} \leq t \leq t_{\max} \quad 0 \leq V \leq 1$$

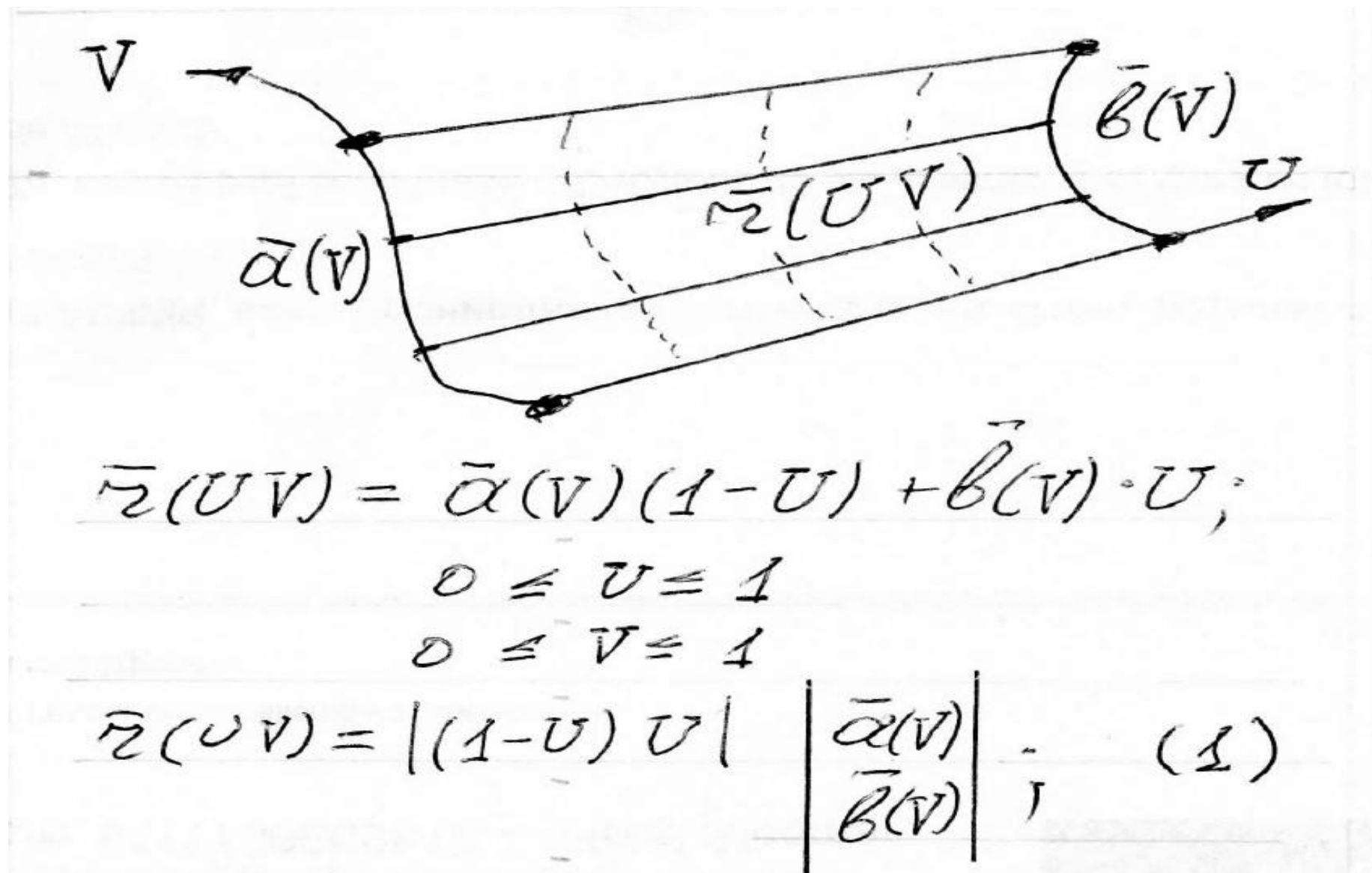
$$z_{\min} \leq z \leq z_{\max}$$

$$t = t_{\min}(1-V) + t_{\max} V;$$

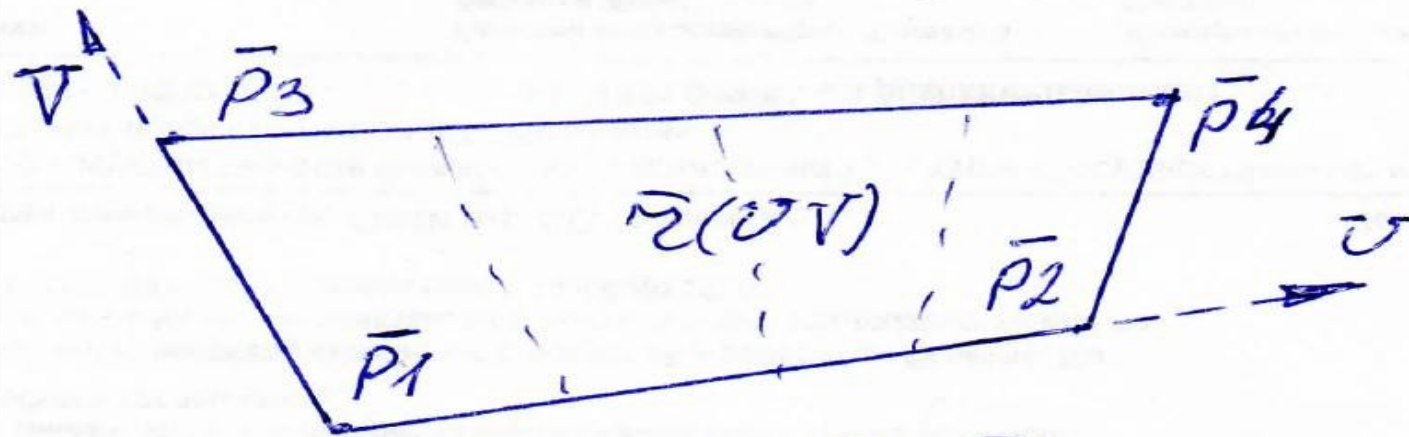
$$z = z_{\min}(1-V) + z_{\max} V;$$

$$0 \leq V \leq 1$$

Линейчатая поверхность



Билинейная поверхность



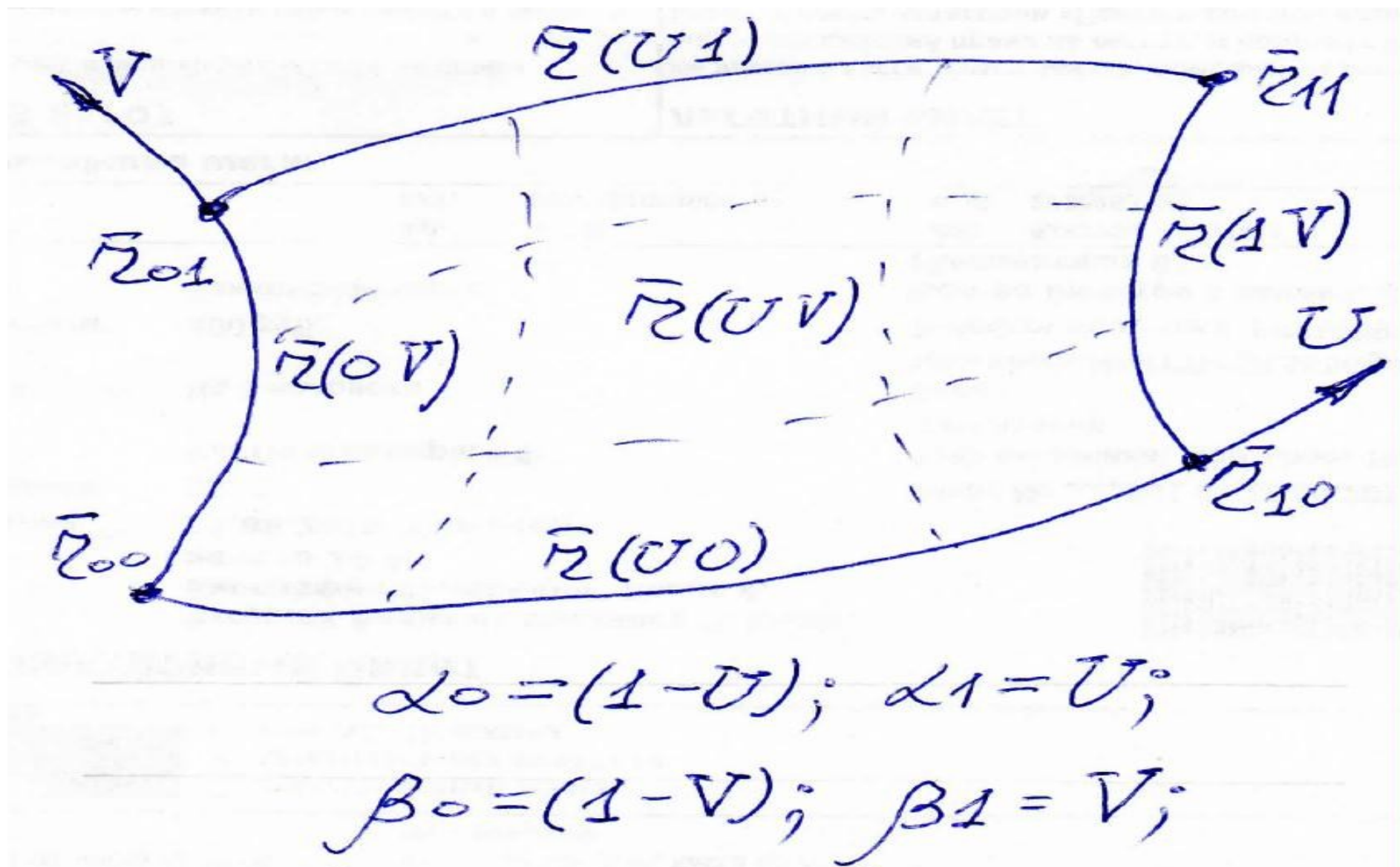
$$\bar{Z}(U, V) = [\bar{P}_1(1-V) + \bar{P}_3 V] \cdot (1-U) + [\bar{P}_2(1-V) + \bar{P}_4 V] \cdot U;$$

$$0 \leq U \leq 1$$

$$0 \leq V \leq 1$$

$$\bar{Z}(U, V) = \begin{vmatrix} (1-V) & V \end{vmatrix} \begin{vmatrix} \bar{P}_1 & \bar{P}_2 \\ \bar{P}_3 & \bar{P}_4 \end{vmatrix} \begin{vmatrix} 1-U \\ U \end{vmatrix}; \quad (2)$$

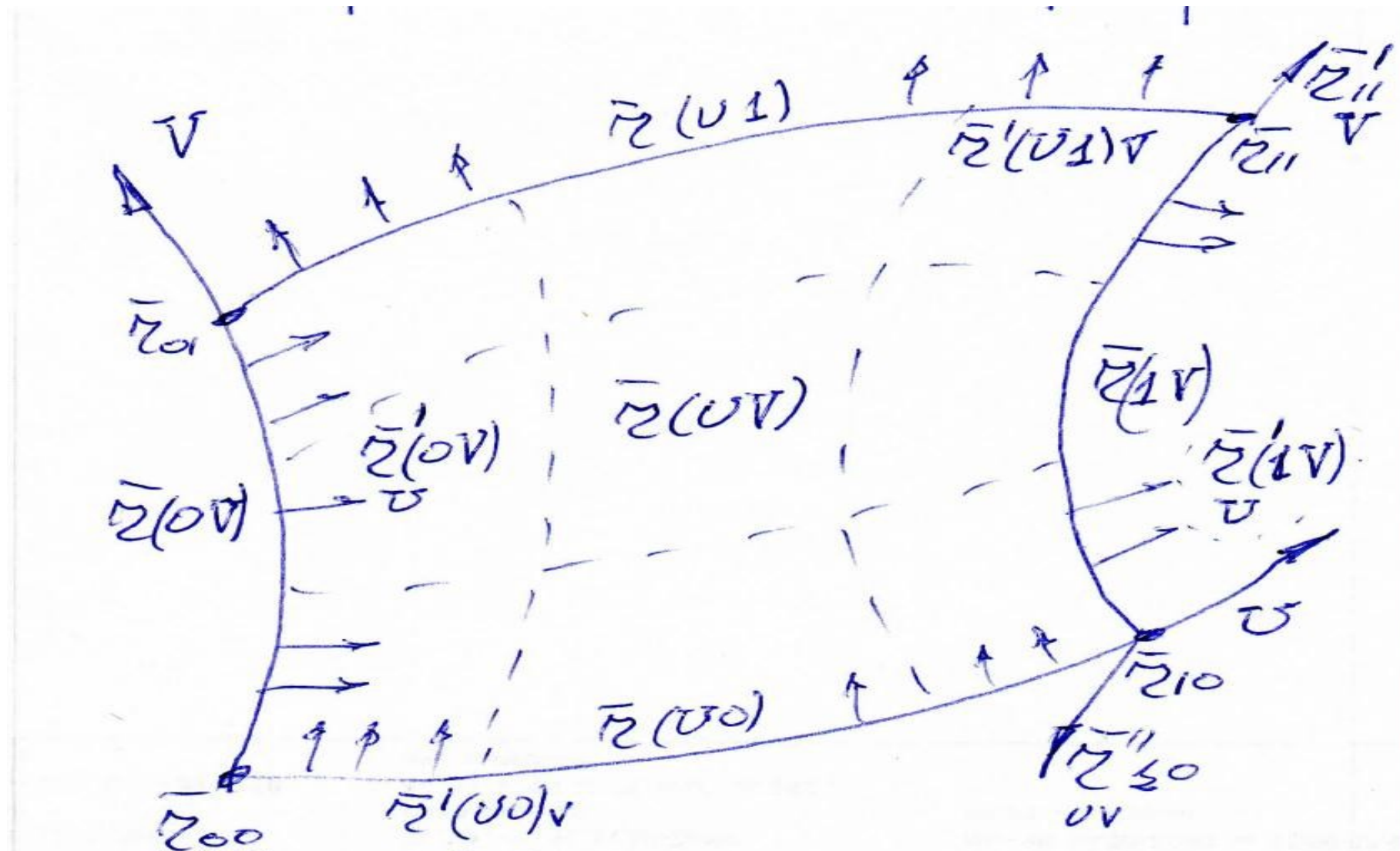
Линейная поверхность Кунса



Формула для линейной поверхности Кунса

$$\begin{aligned}\bar{z}(UV) = & \left| \alpha_0 \alpha_1 \right| \cdot \left| \begin{array}{c} \bar{z}(0V) \\ \bar{z}(1V) \end{array} \right| + \\ & + \left| \beta_0 \beta_1 \right| \cdot \left| \begin{array}{c} \bar{z}(U0) \\ \bar{z}(U1) \end{array} \right| - \\ & - \left| \beta_0 \beta_1 \right| \cdot \left| \begin{array}{cc} \bar{z}_{00} & \bar{z}_{10} \\ \bar{z}_{01} & \bar{z}_{11} \end{array} \right| \cdot \left| \begin{array}{c} \alpha_0 \\ \alpha_1 \end{array} \right| ;\end{aligned}\quad (3)$$

Кубическая поверхность Кунса



Обозначения. Весовые коэффициенты сплайна Эрмита

$$\alpha_0 = 1 - 3U^2 + 2U^3$$

$$\alpha_1 = 3U^2 - 2U^3$$

$$\alpha_2 = U - 2U^2 + U^3$$

$$\alpha_3 = -U^2 + U^3$$

$$\beta_0 = 1 - 3V^2 + 2V^3$$

$$\beta_1 = 3V^2 - 2V^3$$

$$\beta_2 = V - 2V^2 + V^3$$

$$\beta_3 = -V^2 + V^3$$

Формула для кубической поверхности Кунса

$$\begin{aligned}
 \bar{\tau}(U, V) = & \begin{vmatrix} \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 \end{vmatrix} \cdot \begin{vmatrix} \bar{\tau}(0, V) \\ \bar{\tau}(1, V) \\ \bar{\tau}'(0, V)_v \\ \bar{\tau}'(1, V)_v \end{vmatrix} + \\
 & + \begin{vmatrix} \beta_0 & \beta_1 & \beta_2 & \beta_3 \end{vmatrix} \cdot \begin{vmatrix} \bar{\tau}(U, 0) \\ \bar{\tau}(U, 1) \\ \bar{\tau}'(U, 0)_v \\ \bar{\tau}'(U, 1)_v \end{vmatrix} - \\
 & + \begin{vmatrix} \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 \end{vmatrix} \cdot \begin{vmatrix} \bar{\tau}_{00} & \bar{\tau}_{01} & \bar{\tau}'_{00} & \bar{\tau}'_{01} \\ \bar{\tau}_{10} & \bar{\tau}_{11} & \bar{\tau}'_{10} & \bar{\tau}'_{11} \\ \bar{\tau}''_{00} & \bar{\tau}''_{01} & \bar{\tau}''_{00} & \bar{\tau}''_{01} \\ \bar{\tau}''_{10} & \bar{\tau}''_{11} & \bar{\tau}''_{10} & \bar{\tau}''_{11} \end{vmatrix} \begin{vmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{vmatrix}
 \end{aligned}$$